

WASP-52b

R-1181

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_211109 · created 2026-07-31T04:32:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459527.7165 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1822 +/- 0.0089
Transit depth (Rp/Rs)^2	3.32 +/- 0.32 [%]
Orbital Inclination (inc)	85.69 +/- 0.64
Airmass coefficient 1 (a1)	1.273 +/- 0.013
Airmass coefficient 2 (a2)	-0.1723 +/- 0.0061
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	None
Optimal Aperture	4.66
Optimal Annulus	11.99
Transit Duration (day)	0.0788 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1822 $\pm$ 0.0089 vs 0.1646 (deviation 1.91 σ)
Duration fit vs published	0.0788 d vs 0.0758 d (deviation 0.82 σ)
Mid-time O–C	2.8 min
Depth SNR	19.8
Residual scatter	1.02 %

Light curve

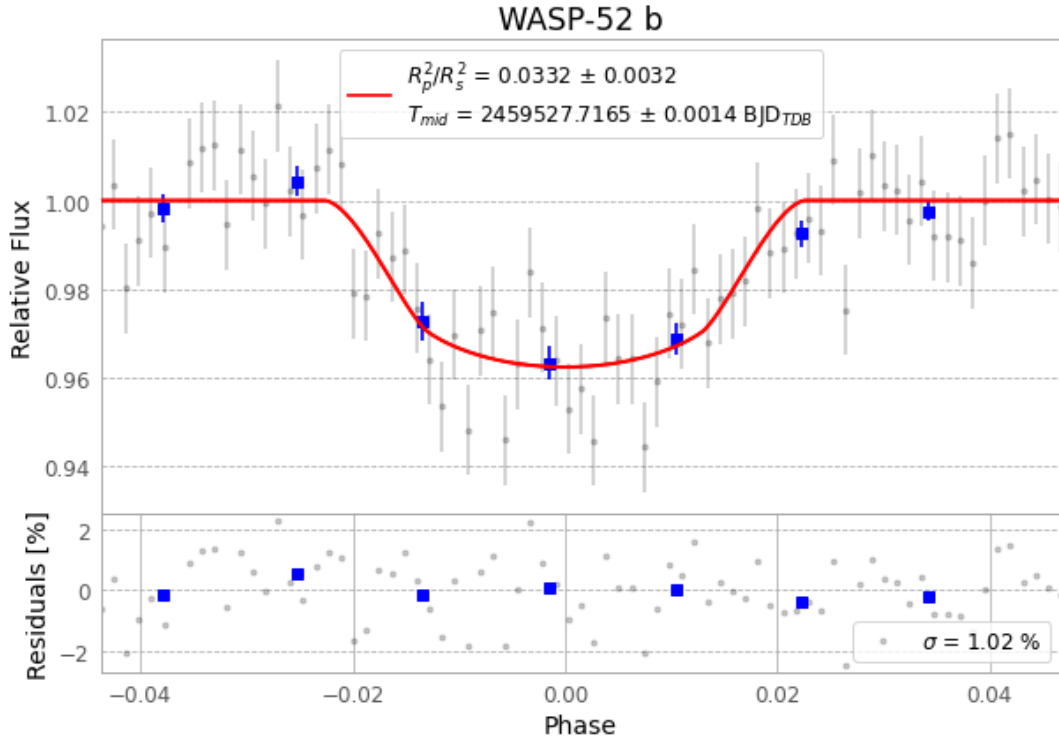


Figure 1 — FinalLightCurve_WASP-52 b_2021-11-08.png (SHA-256 434ff8f13e1e931f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 200], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 07c1f337e02bb080...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52211109031512.FITS → WASP-52211109070312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-211109.log (73c9afa556b9...), FinalLightCurve_WASP-52 b_2021-11-08.png (434ff8f13e1e...), FinalLightCurve_WASP-52 b_2021-11-08.csv (4f7558b4db45...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 04:32Z.